Dynamic Set-Theoretic Normalization of Provenance-Bearing Hyper-Relational Frames

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Abstract

We develop a deterministic semantics for normalizing claims and event frames across a mutable document corpus. The extraction front end may propose several frame interpretations for each claim occurrence, and coreference is not forced to one early clustering. Instead, every exact interpretation and strict coreference partition is a candidate normalization hypothesis. Constraints denote property sets of hypotheses, and the solver is literally set theoretic estimation (STE): it repeatedly intersects its current state with property sets and records the exact set reduction. The resulting feasible family supports a possible-worlds semantics for identity. Coreference holding in every feasible hypothesis defines a safe quotient; disagreement between feasible merge and split hypotheses defines an explicit uncertainty overlay U.

We prove document-insertion antitonicity, deletion expansion, insertion order independence, a delta decomposition by newly activated provenance constraints, identity and composition laws for feasible-world restriction, equivalence of must-coreference, and exact status separation into inconsistent, must, cannot, and uncertain cases. A finite countermodel proves that possible coreference and the uncertainty overlay need not be transitive even though every exact hypothesis is a partition. We distinguish the safe must-quotient from a possible-coreference ambiguity envelope and prove that insertion can strictly split the latter by eliminating a merge world. A two-claim Toyota/BMW model gives a constructive witness and an exact one-candidate reduction. We prove existence of finite minimal document supports for must- and cannot-coreference conclusions. Beyond these dynamic laws, we prove that claim restriction under deletion is necessarily partial, with restriction fibers of size 0 or 1; that gluing feasibility across corpora is controlled by an explicit cross-document constraint obstruction admitting at most one compatible extension; that occurrence-set identifiers give collision-free snapshot identity while cross-snapshot lineage is an intentionally nontransitive relation; that audited typed relations lift to necessary versions one sound rule step at a time; and that dependent downstream truth estimation commutes exactly through its marginal union. The main limitation theorem is sharp: over N ambient worlds, unrestricted crisp STE realizes all 2^N extensional solver states, so no subexponential representation follows from STE alone. A finite executable solver is proved extensionally equal to the abstract semantics. The accompanying Lean 4 development names every stated formal result.

1 Introduction

Cross-document event coreference is usually presented as clustering: a learned model scores pairs or clusters, and an agglomerative or graph algorithm selects one partition. This is an effective predictive architecture, but it is an awkward database semantics. A database receives and retracts documents. A new source can distinguish events previously thought identical; deletion can remove the only evidence that forces a merge or split. Anaphora and omitted arguments can remain genuinely unresolved. A destructive union operation has neither the provenance nor the inverse required by this setting.

Event identity itself is richer than lexical similarity. Work on quasi-identity and dense event annotation distinguishes strict identity from subevent, near-identity, and related-event phenomena [5, 7, 8]. Recent resources make decontextualization, difficult identity boundaries, and framing divergence increasingly explicit [9, 10]. The correct deterministic reaction is not to pretend every ambiguity has a unique answer. It is to retain the admissible answers and state exactly what follows in all, some, or none of them.

This paper formalizes that reaction. We assume a high-recall extraction front end. Each immutable claim occurrence has source provenance and a nonempty set of candidate hyper-relational frames. An exact normalization hypothesis chooses one candidate per claim and supplies an equivalence relation of strict frame identity. Typed nonidentity links—subevent, causation, temporal relation, repeated instance, framing divergence—are conceptually separate and do not participate in the identity quotient.

The contributions are:

1. an STE solver semantics in which property-set application and exact reduction are the complete operational core;
2. a provenance-indexed corpus semantics supporting insertion and deletion without destructive clustering;
3. a must-coreference quotient, a nontransitive uncertainty overlay, and a finite countermodel ruling out possible-coreference clustering;
4. a mechanized insertion-induced ambiguity-envelope split, including a finite cardinal bound and an exact Toyota/BMW witness;
5. contravariant feasible-world restriction, covariant canonical-frame maps, and a necessarily partial claim restriction, exposing the fundamental insertion/deletion asymmetry;
6. an exact cross-document gluing obstruction admitting at most one compatible extension;
7. collision-free snapshot identity from occurrence sets, together with an intentionally nontransitive lineage relation;
8. audited lifting of typed necessary relations, one sound rule step at a time;
9. existence of minimal finite document certificates for settled links;
10. a dependent lift of downstream truth estimation over all feasible normalizations, with exact marginal commutation;
11. a tight 2^N extensional-state bound, together with an executable finite solver proved equal to the abstract semantics.

2 The normalization universe

Fix types of documents, claim occurrences, structured frames, exact normalization hypotheses, and constraints:

$$\mathsf{D}, \qquad \mathsf{C}, \qquad \mathsf{F}, \qquad \Omega, \qquad \mathsf{K}.$$

The Lean structure implementing these data is `STE.DynamicFrame.Model`.

Each claim c has an immutable source document $\mathrm{owner}(c)$ and a nonempty candidate set $\Phi(c) \subseteq \mathsf{F}$. Each exact hypothesis $\omega \in \Omega$ chooses

$$\mathrm{interpret}_\omega(c) \in \Phi(c)$$

and supplies a strict-coreference relation $\sim_\omega$ on claims. The model requires $\sim_\omega$ to be an equivalence relation. Thus every exact hypothesis is an ordinary, logically valid partition. Ambiguity is represented between hypotheses, not by allowing one hypothesis to contain an invalid overlapping "partition."

Every constraint $k \in \mathsf{K}$ has two denotations. First, its provenance support is a set of documents

$$\mathrm{supp}(k) \subseteq \mathsf{D}.$$

Second, it denotes an STE property set

$$S_k = \{\omega \in \Omega \mid \omega \text{ satisfies } k\}.$$

For an active corpus $D \subseteq \mathsf{D}$, the active constraints and feasible normalizations are

$$K(D) = \{k \mid \mathrm{supp}(k) \subseteq D\}, \tag{1}$$
$$\mathcal{N}(D) = \bigcap_{k \in K(D)} S_k. \tag{2}$$

Empty-support logical or ontological constraints are active in every corpus. A constraint derived jointly from several documents is active only while every document in its support remains active.

This fixed ambient universe is deliberate. Documents, claims, constraints, and hypotheses are immutable mathematical objects. Database mutation changes an activation set. Explicit restriction maps for garbage-collecting or forgetting inactive claims exist, but they are necessarily partial (Section 6); semantic retraction does not require erasing the objects that explain prior results.

3 The solver is STE

The phrase "constraint solver" can suggest an additional optimization or clustering semantics. There is none here. The solver state is a set $X \subseteq \Omega$. Applying k is literal STE property-set application:

$$\mathrm{apply}(X, k) = X \cap S_k. \tag{3}$$

The exact reduction caused by k is

$$\mathrm{red}(X, k) = X \setminus (X \cap S_k) = \{\omega \in X \mid \omega \notin S_k\}. \tag{4}$$

These are `Model.applyConstraint` and `Model.solverReduction`.

Proposition 3.1 (One-step STE laws)**.** *Property application is contractive and idempotent; any two property applications commute. Moreover,*

$$\mathrm{apply}(X, k) \cup \mathrm{red}(X, k) = X,$$

and $\mathrm{red}(X, k) = \varnothing$ *exactly when* $X \subseteq S_k$*.*

Lean: applyConstraint_subset, applyConstraint_idempotent, applyConstraint_comm, apply_union_reduction, and solverReduction_eq_empty_iff.

For a finite schedule $[k_1, \ldots, k_n]$, define a run by repeated application, starting from X, and define solving to start from Ω.

Theorem 3.2 (Operational–denotational correspondence). *For every finite list L,*

$$\omega \in \mathrm{run}(L, X) \iff \omega \in X \ \wedge\ \forall k \in L,\ \omega \in S_k, \tag{5}$$

$$\mathrm{solve}(L) = \bigcap_{k \in L} S_k. \tag{6}$$

Consequently, schedules with the same constraint membership have equal output; in particular, permutations and duplicates do not matter.

Lean: mem_run, run_eq_inter_partial, solve_eq_partialFeasibilitySet, solve_congr, and solve_perm.

If a list enumerates exactly $K(D)$, the final operational state equals $\mathcal{N}(D)$ (solve_eq_feasible). Thus a SAT, SMT, ASP, BDD, or custom partition engine is correct only insofar as it implements this meet and its queries. Such technologies are representations and acceleration strategies, not alternate semantics.

Retraction follows the same point. The system does not attempt to invert an old merge. It changes $K(D)$ using provenance and evaluates the new meet. An incremental implementation may reuse caches or truth-maintenance labels, but its extensional correctness criterion is equality with the fresh STE meet. This connects directly to truth-maintenance systems [2, 3], provenance [4], and incremental view maintenance [1].

4 Dynamic corpus laws

Let $D \subseteq E$. Every constraint active in D is active in E, hence

Theorem 4.1 (Information antitonicity).

$$D \subseteq E \quad \Longrightarrow \quad \mathcal{N}(E) \subseteq \mathcal{N}(D).$$

Lean: Model.feasible_antitone. Read forward, document addition removes hypotheses. Read backward, removal can restore hypotheses. The latter is feasible_subset_after_remove.

The elementary database laws are exact:

duplicate insertion is idempotent (feasible_insert_idempotent);

insertion order is irrelevant (feasible_insert_comm);

if $d \notin D$, adding and then removing d restores precisely $\mathcal{N}(D)$ (feasible_insert_then_remove).

Define the constraint delta

$$\Delta K(D, E) = K(E) \setminus K(D).$$

Then addition has an exact incremental decomposition.

Theorem 4.2 (Delta decomposition). *If $D \subseteq E$, then*

$$\omega \in \mathcal{N}(E) \iff \omega \in \mathcal{N}(D) \ \wedge\ \forall k \in \Delta K(D, E),\ \omega \in S_k.$$

Lean: `mem_feasible_extension`. If the delta is empty, the feasible sets are equal (`feasible_eq_of_no_new_constraints`).

Constraint activation interacts asymmetrically with union and intersection:

$$K(D \cap E) = K(D) \cap K(E), \qquad K(D) \cup K(E) \subseteq K(D \cup E).$$

The second inclusion can be strict because $D \cup E$ may activate a constraint whose support draws documents from both sides. Lean: `activeConstraints_inter` and `activeConstraints_union_subset`. This is the first genuinely cross-document effect in the theory; Section 7 shows that it is exactly the obstruction to gluing feasibility across corpora.

5 Must, may, cannot, and uncertainty

For $c, d \in \mathsf{C}$ and a consistent corpus, define

$$\begin{aligned} \mathsf{Must}_D(c,d) &\iff \forall \omega \in \mathcal{N}(D),\ c \sim_\omega d, &(7)\\ \mathsf{May}_D(c,d) &\iff \exists \omega \in \mathcal{N}(D),\ c \sim_\omega d, &(8)\\ \mathsf{Cannot}_D(c,d) &\iff \forall \omega \in \mathcal{N}(D),\ c \not\sim_\omega d, &(9)\\ \mathsf{U}_D(c,d) &\iff (\exists \omega \in \mathcal{N}(D),\ c \sim_\omega d) \wedge (\exists \omega \in \mathcal{N}(D),\ c \not\sim_\omega d). &(10) \end{aligned}$$

This is the certain/possible distinction of incomplete-database possible worlds [6] applied to normalization partitions.

Consistency must be checked first. If $\mathcal{N}(D) = \varnothing$, both universal statements are vacuously true. The implemented status type therefore has four constructors:

$$\mathsf{inconsistent}, \quad \mathsf{must}, \quad \mathsf{cannot}, \quad \mathsf{uncertain}.$$

Lean: `Model.CorefStatus` and `Model.corefStatus`, with one correctness theorem for each constructor.

Theorem 5.1 (Safe quotient)**.** Must_D *is an equivalence relation. Therefore the deterministic normalized frame collection*

$$F'_D = \mathsf{C}(D)/\mathsf{Must}_D$$

is well defined on active claims.

Proof. Reflexivity, symmetry, and transitivity hold in every exact relation $\sim_\omega$ and are preserved by intersection over $\omega \in \mathcal{N}(D)$. □

Lean: `mustSame_equivalence`, `activeMustSetoid`, and `CanonicalFrame`.

Possible coreference is reflexive on a consistent corpus and symmetric, but no general transitivity theorem is available. U_D is symmetric and irreflexive. Must and cannot are each disjoint from uncertainty. Lean: `maySame_refl`, `maySame_symm`, `uncertain_symm`, `uncertain_irrefl`, `must_not_uncertain`, and `cannot_not_uncertain`.

5.1 Status evolution under corpus change

The modal laws give a concise transition discipline. Under $D \subseteq E$:

$$\begin{aligned} \mathsf{Must}_D(c,d) &\Longrightarrow \mathsf{Must}_E(c,d), \\ \mathsf{Cannot}_D(c,d) &\Longrightarrow \mathsf{Cannot}_E(c,d), \\ \mathsf{May}_E(c,d) &\Longrightarrow \mathsf{May}_D(c,d), \\ \mathsf{U}_E(c,d) &\Longrightarrow \mathsf{U}_D(c,d). \end{aligned}$$

Thus addition can resolve uncertainty into must or cannot, or make the entire problem inconsistent. It cannot turn a consistent settled fact into uncertainty within the hard-constraint semantics. Deletion reverses the information order: a settled link may become uncertain, but a possibility cannot be invented by adding a constraint.

These are `mustSame_mono`, `cannotSame_mono`, `maySame_antitone`, and `uncertain_antitone`. They are useful both as theorems and as property-based tests for an incremental implementation.

5.2 A finite countermodel

Let the mentions be a, b, c and let two exact worlds be feasible. The left world has blocks $\{a, b\}, \{c\}$; the right world has blocks $\{a\}, \{b, c\}$. Each world is an ordinary partition. Nevertheless,

$$\mathsf{May}(a,b), \qquad \mathsf{May}(b,c), \qquad \neg\mathsf{May}(a,c).$$

Moreover (a, b) and (b, c) are uncertain while (a, c) is cannot-corefer.

Theorem 5.2 (Nontransitivity)**.** *Neither* May *nor* U *is transitive in general.*

The model and proofs are fully finite and constructive in `Ste.DynamicFrame.Counterexample`; the theorem names are `maySame_not_transitive` and `uncertain_not_transitive`.

This establishes a design constraint. U is an overlay between must-classes, not a special kind of class. Taking connected components of possible or uncertain edges invents identifications not licensed in any exact world.

5.3 Insertion can split an ambiguity envelope

The preceding warning does not mean that an existing database-facing frame can never split. It means that we must say which object bears the word "frame." Define the possible-coreference envelope anchored at c by

$$A_D(c) = \{d \in \mathsf{C} : \mathsf{May}_D(c,d)\}.$$

This is a presentation of unresolved alternatives, not an equivalence class. For $D \subseteq E$, feasibility antitonicity gives

$$A_E(c) \subseteq A_D(c).$$

The inclusion can be strict. Say that insertion *resolves* c, d *apart* when

$$\mathsf{U}_D(c,d) \quad \text{and} \quad \mathsf{Cannot}_E(c,d).$$

Theorem 5.3 (Insertion-induced envelope splitting)**.** *If* $D \subseteq E$ *resolves* c, d *apart, then*

$$A_E(c) \subseteq A_D(c), \qquad d \in A_D(c), \qquad d \notin A_E(c).$$

If the claim universe is finite, then

$$|A_E(c)| < |A_D(c)| \le |\mathsf{C}|.$$

Thus the exact envelope reduction $|A_D(c)| - |A_E(c)|$ *is positive and is at most* $|\mathsf{C}|$ *without further assumptions.*

Proof. The old uncertainty supplies a feasible world merging c and d, hence $d \in A_D(c)$. The new cannot-link is equivalent to absence of every feasible merge world, hence $d \notin A_E(c)$. Antitonicity of May supplies the subset. For finite sets, a proper subset has strictly smaller cardinality; both envelopes are subsets of the ambient claim universe. □

Lean: `Model.mayEnvelope_antitone` and `Model.mayEnvelope_split_witness`.
The finite theorem names are

`Model.finiteMayEnvelope_card_lt_of_resolvesApart`
`Model.envelopeReduction_le_card`.

The smallest informative witness contains two old descriptions and two exact worlds. Initially c is "the red car" and d is "the burgundy car." One world treats them as one vehicle and the other as two, so

$$\mathcal{N}(D) = \{\{\{c,d\}\},\ \{\{c\},\{d\}\}\}\,.$$

A later document identifies the red description as Toyota and the burgundy description as BMW. Under the audited property that one vehicle cannot have both manufacturers, STE rejects the one-vehicle world:

$$\mathcal{N}(E) = \{\{\{c\},\{d\}\}\}\,.$$

Consequently $A_D(c) = \{c,d\}$, $A_E(c) = \{c\}$, and the exact reduction is 1. This construction is mechanized in module `Ste.DynamicFrame.AmbiguitySplit`, namespace `VehicleSplit`; the principal results are `toyota_bmw_envelope_split` and `toyota_bmw_split_into_two_singletons`, with the exact reduction certified by `toyota_bmw_reduction_eq_one`.

There is no contradiction with monotonicity of the must-quotient. If $\mathsf{Must}_D(c,d)$ holds, constraint intersection cannot make it false: every later world is already an earlier world. The split occurs because the old database-facing object is an ambiguity envelope containing a *possible* link, not a forced identity class. The authoritative state is therefore the feasible family of exact partitions. A single partition plus pairwise U marks is only a summary and can lose correlations among alternatives.

6 Restriction, quotient maps, and retraction asymmetry

On the fixed ambient hypothesis universe, inclusion $D \subseteq E$ induces

$$\rho_{E,D} : \mathcal{N}(E) \longrightarrow \mathcal{N}(D), \qquad \rho_{E,D}(\omega) = \omega.$$

This is a genuine restriction map: it forgets the additional requirements, not the underlying hypothesis.

Theorem 6.1 (Restriction laws)**.** *Restriction by identity is identity, and for* $D \subseteq E \subseteq F$*,*

$$\rho_{E,D} \circ \rho_{F,E} = \rho_{F,D}.$$

A point of $\mathcal{N}(D)$ *lies in the image of* $\rho_{E,D}$ *exactly when the same ambient hypothesis is feasible in* E*.*

Lean: `restrictFeasible_id`, `restrictFeasible_comp`, and `mem_range_restrictFeasible_iff`. Writing $W(D) = \mathcal{N}(D)$ for the subtype of exact worlds feasible at corpus D, the same identity and composition laws hold verbatim for the packaged subtype maps (`restrictWorld_id` and `restrictWorld_comp`), so feasible worlds form a contravariant functor on the document-set poset without any new axiom.

Thus $D \mapsto \mathcal{N}(D)$ is already a contravariant set-valued construction on the poset of active document sets. Calling it a presheaf becomes literal once the corpus-dependent universe and a target category are fixed. Calling it a sheaf requires a further gluing theorem; Section 7 isolates the exact obstruction.

The quotient moves the other way. Since

$$\mathsf{Must}_D(c, d) \Longrightarrow \mathsf{Must}_E(c, d),$$

old must-classes map into new must-classes. Active-claim inclusion and relation preservation induce

$$q_{D,E} : F'_D \longrightarrow F'_E.$$

Lean: `includeActiveClaim` and `canonicalMap`. This map need not be injective: added evidence can eliminate all split worlds and thereby merge old must-classes. It also need not be surjective because E contains new active claims.

There is generally no quotient map in the deletion direction. A pair forced together in E may become uncertain in D after the supporting document is removed. Mapping one E-class back would then require splitting it, which is not a function on quotient classes. This is the formal reason that deletion cannot be implemented as the inverse of an agglomerative merge history.

6.1 Claim restriction is necessarily partial

The world level restricts totally; the claim level does not. Insertion gives a total map $\mathsf{C}(D) \to \mathsf{C}(E)$, but deletion cannot give a total reverse map: a claim whose source lies in $E \setminus D$ has no active target. Define a partial restriction relation by equality of immutable occurrence identifiers.

Theorem 6.2 (Partial claim restriction)**.** *For $D \subseteq E$ and $c \in \mathsf{C}(E)$, a restriction of c to $\mathsf{C}(D)$ exists exactly when the source document of c lies in D; when it exists, it is unique.*

Proof. If a restricted active claim is equal to c, its source is active in D. Conversely, source membership constructs the subtype element directly. Equality of two such subtype elements follows from equality of their immutable occurrences. □

Lean: `claimRestriction_exists_iff` and `claimRestriction_unique`. The exact fiber constant is therefore 0 or 1, not always 1; the partiality is intrinsic to variable-universe retraction, not an artifact of the mechanization. Local strict-coreference is preserved definitionally by restricting the world and including the surviving claims (`localSame_restrict`). This is the precise variable-universe result: worlds restrict totally and functorially, while claims restrict partially with unique values wherever a value exists.

7 Gluing feasibility across corpora

Restriction moves down the document poset one corpus at a time. Gluing asks the transverse question: when do hypotheses feasible on two corpora assemble into a hypothesis feasible on their union? Section 4 shows that the inclusion $K(D) \cup K(E) \subseteq K(D \cup E)$ can be strict; the strictness is carried by constraints whose support straddles both corpora, and exactly those constraints obstruct gluing. For an ambient hypothesis h, define the obstruction

$$\mathcal{O}_{D,E}(h) = \{k \in \mathsf{K} : \mathrm{supp}(k) \subseteq D \cup E, \\ \mathrm{supp}(k) \nsubseteq D, \mathrm{supp}(k) \nsubseteq E, h \notin S_k\}.$$

Only genuinely cross-document constraints occur in this set.

Theorem 7.1 (Exact union obstruction)**.** *For every h,*

$$h \in \mathcal{N}(D \cup E) \iff h \in \mathcal{N}(D) \wedge h \in \mathcal{N}(E) \wedge \mathcal{O}_{D,E}(h) = \varnothing.$$

Proof. The forward direction follows by feasibility antitonicity and satisfaction of every union-active constraint. Conversely, a union-active constraint is active on D, active on E, or active on neither side. The first two cases follow from local feasibility. In the third case, violation would place the constraint in $\mathcal{O}_{D,E}(h)$, contradicting emptiness. □

Lean: `mem_feasible_union_iff`.

Two local views are called compatible here when they expose the same ambient exact hypothesis. Under that qualification, the global-extension type is nonempty exactly when the obstruction vanishes (`globalExtension_nonempty_iff`), and it is a subsingleton (`globalExtension_subsingleton`). Hence the number of extensions is at most 1. For finite K with decidable support and satisfaction, the obstruction is computable by one filter and contains at most $|\mathsf{K}|$ elements. If compatibility is weakened to agreement only on overlap, uniqueness is no longer promised: residual identity choices outside the overlap are precisely additional global worlds.

8 Stable snapshot identity and relational lineage

A database publishes the must-classes of the covariant quotient, so it must name them. Identifiers generated at cluster-creation time break under merge and split; the correct identifier is the member set itself. For a canonical must-class q at snapshot D, define

$$\mathrm{id}_D(q) = \{c \in \mathsf{C}(D) : c \text{ belongs to } q\}.$$

These are immutable claim occurrence identifiers, not generated cluster names.

Theorem 8.1 (Complete snapshot identifier)**.** *Every $\mathrm{id}_D(q)$ is nonempty, and $\mathrm{id}_D(q) = \mathrm{id}_D(r)$ implies $q = r$. For $D \subseteq E$,*

$$\mathrm{id}_D(q) \subseteq \mathrm{id}_E(q_{D,E}(q)).$$

Proof. A representative belongs to its own equivalence class. Equal member sets put each representative in the other's class, so quotient soundness gives equal classes. Under insertion, claims remain active and must-coreference is monotone, proving the inclusion. □

Lean: `canonicalMembers_nonempty`, `canonicalMembers_injective`, and `canonicalMembers_subset_map`. Thus the minimum identifier size is 1, and the identifier has zero collisions inside a snapshot.

Across arbitrary snapshots define lineage by nonempty intersection of member sets. It is reflexive for live frames, symmetric, and every frame has lineage to its insertion image (`lineage_refl`, `lineage_symm`, and `lineage_canonicalMap`). It is intentionally not transitive: the sets $\{a\}$, $\{a, b\}$, and $\{b\}$ give the smallest counterexample. Lean proves this on a two-element type in `overlap_not_transitive`. Consequently a single immutable scalar identifier cannot simultaneously model a merge and both later split children without collision or arbitrary choice. Stable snapshot identity plus relational lineage is achievable; globally functional identity through splits is not.

9 Audited lifting of typed relations

Typed nonidentity links—subevent, causation, temporal relation, repeated instance, framing divergence—are deliberately excluded from the identity quotient, but they too hold or fail world by world, so each has a must-version obtained by intersecting over feasible worlds. The question is which inference rules survive the intersection.

The mechanization gives each relation kind a source and target sort and keeps strict identity, subevent, supersession, before, causes, repeated instance, and framing divergence distinct. A concrete world semantics supplies the laws it actually validates: transitivity for subevent, supersession, and before; causation implying temporal precedence; equivalence laws for repeated instance; and symmetry plus irreflexivity for framing divergence.

Theorem 9.1 (Necessary-relation lifting)**.** *Let a composition rule*

$$R_i(x, y) \wedge R_j(y, z) \Longrightarrow R_k(x, z)$$

hold in every exact world. Replacing each relation by its must-version—the intersection over feasible worlds—preserves the same rule.

Proof. Fix a feasible world h. Both must-premises hold in h, so world-level soundness gives the conclusion in h. Since h is arbitrary, the must-conclusion follows. □

Lean: `CompositionRule` and `must_compose`. Specialized proofs are `mustSubevent_trans`, `mustSupersession_trans`, `mustBefore_trans`, and `mustCausesBefore`; the repeated-instance and framing-divergence laws are lifted separately. The proof preserves one rule application exactly: two necessary premises produce one necessary conclusion. No theorem asserts, for example, transitivity of causation or framing divergence. Names do not license composition constants; audited world laws do.

Consistency is required to recover endpoint data from a necessary statement, because universal relations over an empty feasible set are vacuous. This is explicit in `mustRel_typed`, `mustStrictIdentity_iff`, and `not_mustDivergence_self`.

10 Minimal document supports

An explanation should identify documents sufficient for a settled conclusion. For a finite document set D, say D supports a must-link when $\mathcal{N}(D)$ is nonempty and $\mathsf{Must}_D(c, d)$ holds. Define cannot-support analogously. A support is minimal when no proper subset still supports the conclusion.

Theorem 10.1 (Finite minimal certificates)**.** *If a finite corpus supports a must-coreference conclusion, it contains an inclusion-minimal finite support for that conclusion. The same holds for a cannot-coreference conclusion.*

Proof. Among the finite supporting subsets of the original corpus, choose one of minimum cardinality. Every proper subset has smaller cardinality and therefore cannot support the conclusion. □

Lean: the generic finite minimization theorem is `exists_minimal_subfinset`; its two applications are `exists_minimal_mustSupport` and `exists_minimal_cannotSupport`.

The theorem is semantic, not an efficiency claim. Computing all minimal supports may be exponential. ATMS labels, unsatisfiable cores, provenance polynomials, and hitting-set dualization are candidate representations. The proof does establish a precise contract for an explanation engine: a returned certificate should be sufficient, consistent, and inclusion-minimal.

11 Lifting truth over normalization worlds

Let T be a type of downstream truth worlds and let

$$T : \Omega \longrightarrow \mathcal{P}(\mathsf{T})$$

assign a truth feasible set to each exact normalization. The joint feasible set is

$$J(D) = \{(\omega, t) \mid \omega \in \mathcal{N}(D),\ t \in T(\omega)\}.$$

This is `TruthLift.jointFeasible`. A property $P(\omega, t)$ is certain when it holds throughout $J(D)$ and possible when it holds somewhere in $J(D)$.

Theorem 11.1 (Truth-lift monotonicity)**.** *If $D \subseteq E$, every property certain over $J(D)$ is certain over $J(E)$, and every property possible over $J(E)$ is already possible over $J(D)$.*

Lean: `certain_mono` and `possible_antitone`.

Not every feasible normalization need support a downstream truth world. The projection of $J(D)$ onto normalization hypotheses is contained in $\mathcal{N}(D)$, with equality if every feasible normalization has a nonempty truth fiber. Lean: `normalizationProjection_subset` and `normalizationProjection_eq`.

The first commutation result covers the clean independence case.

Theorem 11.2 (Normalization-independent commutation)**.** *Assume $\mathcal{N}(D) \neq \varnothing$, $T(\omega) = T_0$ for every ω, and $P(\omega, t) = P_0(t)$. Then*

$$\forall(\omega, t) \in J(D),\ P_0(t) \Longleftrightarrow \forall t \in T_0,\ P_0(t).$$

Existentially, joint possibility is equivalent to consistency of $\mathcal{N}(D)$ together with ordinary possibility in T_0.

Lean: `certain_constant_iff` and `possible_constant_iff`.

This identifies the exact modularity boundary. Normalization-first inference commutes automatically when truth semantics ignores normalization. The independence assumption weakens in two exact directions.

11.1 Dependent fibers commute through the marginal union

For dependent truth fibers $T(\omega)$ define the marginal union

$$T_{\exists}(D) = \{t : \exists\omega \in \mathcal{N}(D),\ t \in T(\omega)\}.$$

The fibers may vary arbitrarily.

Theorem 11.3 (Marginal commutation)**.** *For every truth-only predicate Q,*

$$\forall\omega \in \mathcal{N}(D)\ \forall t \in T(\omega),\ Q(t) \Longleftrightarrow \forall t \in T_{\exists}(D),\ Q(t),$$
$$\exists\omega \in \mathcal{N}(D)\ \exists t \in T(\omega),\ Q(t) \Longleftrightarrow \exists t \in T_{\exists}(D),\ Q(t).$$

Proof. Both equivalences follow by unpacking the existential witness in the definition of $T_{\exists}(D)$. No fiber constancy or nonemptiness premise is used. □

Lean: `certain_marginal_iff` and `possible_marginal_iff`.

A genuinely dependent query $P(\omega, t)$ also commutes if, on jointly feasible support, there is a $Q(t)$ with $P(\omega, t) \leftrightarrow Q(t)$. This checkable condition is `SupportedInvariant`; the exact universal and existential results are `certain_supportedInvariant_iff` and `possible_supportedInvariant_iff`. The loss constant is zero: these are logical equivalences, not approximations. The condition is also meaningful. Two feasible normalizations sharing one truth world but disagreeing on P rule out every truth-only factor Q; this two-normalization, one-truth-world witness is formalized by `no_supportedInvariant_of_disagreement`. Together the marginal and supported-invariant results keep normalization uncertainty intact downstream without enumerating every joint world.

12 The tight exponential limit

The results so far are uniformly positive under explicit qualifications, and every one of them is extensional: a solver state is a subset of Ω, and correctness of any engine means equality with an STE meet. It is natural to hope that the reachable states admit a uniformly compact representation. The following theorem—the paper's principal limitation result—shows that no such representation follows from STE alone, and that the failure is not an artifact of proof technique: the exponential bound is tight.

Assume the ambient exact hypothesis universe is finite of cardinality

$$N = |\Omega|.$$

Theorem 12.1 (Tight extensional state bound)**.** *There are exactly 2^N extensional solver states over N ambient worlds, and unrestricted crisp STE can realize every one of them with a single property set.*

Proof. The solver states are the powerset of the N-element universe, whose cardinality is 2^N. Given any $A \subseteq \Omega$, use one constraint with property set $S_\star = A$; its feasibility set is exactly A. □

Lean: `card_allExactStates` and `arbitrary_subset_as_single_property`.

Therefore no uniform subexponential extensional representation follows from STE alone. BDDs, decomposable circuits, treewidth bounds, or factorized partition structures remain worthwhile, but their compactness theorems must assume structure in the generated property sets. This bound delimits what any implementation, including the executable solver of Section 13, can promise unconditionally.

13 Finite implementation and its correctness contract

The limitation theorem bounds the worst case; it does not impede an exact finite implementation. Assume, as in Section 12, a finite ambient universe with decidable satisfaction. Represent a solver state by a finite set and implement property application by filtering.

Theorem 13.1 (Executable exactness)**.** *Finite filtering denotes exactly abstract STE intersection, and a finite run denotes exactly the corresponding abstract run and partial feasibility set.*

Proof. Membership after filtering is membership in the input conjoined with satisfaction. Induction on the constraint list then identifies every finite run with repeated property-set intersection. □

Lean: `coe_applyFinite`, `coe_runFinite`, and `coe_solveFinite`.

Every nontrivial application removes at least one world (`applyFinite_card_lt`). Total reduction from any state is at most N (`reductionCost_le_maxWorlds`). These are useful operational constants: the explicit representation stores at most N survivors, a full must/may scan examines at most N worlds, and no execution can eliminate more than N distinct worlds.

Beyond these bounds, the engineering surface retains exact correctness oracles: incremental output must equal a fresh STE meet; insertion order must not matter; deletion must reactivate every world excluded only by removed support; and every published settled link should admit a finite minimal certificate when the active corpus is finite.

13.1 Interfaces to learned coreference

Nothing in the theory forbids neural models. They can propose candidate frame interpretations, candidate links, and likely high-reduction criteria. They can rank feasible worlds. What they cannot do silently is turn an uncalibrated score threshold into a semantic impossibility.

This yields three distinct interfaces:

1. *generation*: place defensible worlds inside the ambient candidate universe;
2. *hard admissibility*: apply audited property sets by STE;
3. *ranking*: order the worlds that remain.

Candidate recall is a fairness condition on generation. Hard-constraint soundness is a fairness condition on property sets. Ranking does not alter must/cannot guarantees unless the ranked tail is explicitly discarded, in which case that discard is itself a property-set application that must be named and audited.

13.2 Mechanization map

The development is split by responsibility:

Lean module	Formal content
`Ste.DynamicFrame`	model, feasibility, modal relations, quotient
`Ste.DynamicFrame.Laws`	dynamic laws, restriction, status, quotient map
`Ste.DynamicFrame.Solver`	property application, reduction, finite runs
`Ste.DynamicFrame.Counterexample`	three-mention nontransitivity model
`Ste.DynamicFrame.Support`	finite minimal document certificates
`Ste.DynamicFrame.PresheafGluing`	partial restriction and exact gluing
`Ste.DynamicFrame.FiniteSolver`	executable solver and tight bounds
`Ste.DynamicFrame.RelationAlgebra`	audited typed necessary relations
`Ste.DynamicFrame.TruthLift`	dependent downstream feasible family
`Ste.DynamicFrame.Lineage`	occurrence-set identifiers and lineage
`Ste.DynamicFrame.AmbiguitySplit`	insertion-induced envelope splitting

The root module `Ste.lean` imports the complete theory. The project is pinned to Lean 4.32.0 and Mathlib 4.32.0. The repository workflow builds the root library on pushes to the development branch. The source and this paper use a strict convention: a named Lean theorem is the authoritative statement.

14 Summary of constants and limits

Each thematic section establishes a sharp constant or an explicit limit; the table collects them in the order of the body.

Result	Constant	Qualification or sharp limit
Ambiguity-envelope split	reduction ≥ 1; at most $\lvert\mathsf{C}\rvert$	old may-link eliminated by evidence
Claim restriction	fiber size 0 or 1	source must remain active
Compatible gluing	extensions ≤ 1	locals name the same ambient world
Snapshot identity	size ≥ 1; zero collisions	history is overlap, not a function
Typed composition	one sound rule step	only audited rules lift
Truth commutation	zero logical loss	invariant on feasible support
Finite exact solver	N survivors; 2^N states	exponential bound is tight

15 Conclusion

Dynamic frame normalization is best understood as set-theoretic estimation over exact partitions, not as an irreversible clustering task. Property-set application is the solver. Provenance selects the active meet. Must coreference is the intersection of feasible equivalence relations and supplies the only generally safe deterministic quotient. 'May' and U are modal summaries, and a finite machine-checked countermodel shows why they cannot be collapsed into partitions. Nevertheless, a database-facing ambiguity envelope can genuinely split under insertion: new properties remove merge worlds, shrink May, and can force formerly compatible descriptions apart. The Toyota/BMW witness makes this distinction constructive. Document addition and deletion act in opposite information directions: feasible worlds restrict contravariantly, canonical quotients map covariantly under addition, and claim restriction under deletion is irreducibly partial. Cross-document gluing has an exact obstruction, snapshot identity by occurrence sets is collision-free while lineage is an intentionally nontransitive relation, typed necessary relations compose only under audited world laws, minimal finite supports provide an explanation contract, and the dependent truth lift keeps normalization uncertainty intact downstream.

The result is a substantive mathematical kernel for a live document database: ambiguity is represented, reduction is exact, retraction has a denotation, cross-document gluing has an obstruction, and finite implementation has both an executable correctness theorem and a tight worst-case bound. The 2^N state bound delimits the theory honestly: every structural question raised by the dynamic semantics is settled here either by a proof under explicit hypotheses or by a machine-checked counterexample, and no subexponential extensional representation follows from STE alone. The principal remaining algorithmic question is therefore structural rather than semantic—which properties of real candidate graphs make the exact 2^N worst case compress in practice while retaining deletion, explanation, and must/may queries.

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